

The Science Communication Paradox: Is "It Depends" a Story Worth Telling?

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When Science Communicates, But No One Listens

Science thrives on uncertainty. But science communication can drown in it. We aim to inform, not persuade; to acknowledge complexity, not oversimplify. But if every answer begins and ends with "it depends," are we truly enlightening—or just losing our audience?

Morning light spills through the café window, catching on my laptop screen— "Faktencheck Artenvielfalt: Bodenbiodiversität"¹ lies open. I squint at the cautious phrases—but all I feel is the chill of my cold coffee, mirroring my fading motivation.

Soil biodiversity—finally, a topic that matters! Yet the words feel as lifeless as compacted dirt. I should be out there, hands in the soil, watching beetles scuttle under decomposing leaves, sensing the pulse of microbial life. Instead, I'm here, buried in academia, surrounded by exhausted students who'd rather dive head-first into a swamp than explain why recycling alone won't save the planet. There's an unspoken frustration hanging thick in the air, and I know we share the same itch beneath our skin: we'd rather show than tell—but grandma or the local government is hardly likely to follow us into the field.

As I flip through the chapter, a question gnaws at me: How do we explain without oversimplifying? Guide without overpromising? The underground world is an intricate, living system—yet here, it's reduced to carefully hedged statements, where every claim is buffered by uncertainty. The scientist in me understands. The communicator in me sighs.

Does this chapter do science communication wrong? Or is it playing a game it cannot win?

The Strength: Communicating Uncertainty Well (Maybe Too Well?)

The chapter gets one thing right: it acknowledges the vast uncertainties in soil biodiversity research. It repeats the same refrain: The data are incomplete, the studies are few, the gaps are vast.

"More than 60% of European soils are damaged. However, we cannot even roughly estimate the impact of this degradation on soil biodiversity."^a

^aOriginal German: "Mehr als 60 % der europäischen Böden sind geschädigt. Welche Auswirkungen diese Schädigung auf die Bodenbiodiversität hat, können wir dagegen nicht einmal grob einschätzen." (p. 918)

Honest. Transparent. This is science. But transparency alone isn't enough. Without a *guiding narrative*, the chapter leaves readers floating, with no foothold in what is known. If uncertainty is all we highlight, do we risk making science seem directionless?

Instead of merely stating what we don't know, science communication should provide a **roadmap forward**:

"While long-term studies are still lacking, evidence from X and Y suggests soil biodiversity plays a critical role in climate resilience. Research is exploring A and B to fill this gap."

We don't have to pretend to know everything—but shouldn't we show that science is actively working toward answers?

I sip my coffee and turn another page. So far so good — known terrain. But the headline of the next chapter jumps at me: "Handlungsoptionen" ("Call to Action"). We are moving from the firm ground of scientific knowledge to the thin ice of public guidance, thus communicating not to other scientists but to people outside — people with power. The transition is stark: a shift from what we don't know to what we must do. But can we make recommendations without overstating certainty? Can we stay neutral while still guiding action?

The Caution Trap: Inform, Not Persuade—But How?

Science communication teaches us to *inform, not persuade*. Facts should be laid out clearly, options presented neutrally, and audiences trusted to make their own decisions. But when it comes to *Handlungsoptionen* ("Actionable Options")—the part of the chapter outlining what should be done—this principle starts to wobble.

¹[Eisenhauer et al., 2023]

Policy makers, industry leaders, land managers—those with power to act—don't sift through pages of uncertainty before making decisions. Or do they?

One strength of the chapter is presenting options without imposing an agenda. Instead of dictating a path, it provides a menu of possibilities, allowing flexibility depending on context.

"The choice of specific measures should align with local conditions, considering both natural factors and the economic and organizational realities of an operation."^a

^aOriginal German: "Dabei sollte die konkrete Maßnahmenwahl den lokalen Bedingungen entsprechen, sowohl was die natürlichen Gegebenheiten als auch ggfs. die wirtschaftlichen und organisatorischen Realitäten eines Betriebs betrifft." (p. 918)

This is effective science communication—acknowledging complexity while still offering direction. There is no single solution to improving soil biodiversity, and this wording respects that.

Similarly, when addressing landowners:

"Landowners who do not cultivate their own land (e.g., lessors) have the option to require biodiversity-friendly practices or restrict particularly harmful ones on their property."^a

^aOriginal German: "Bodeneigentümer:innen, die ihren Boden nicht selbst nutzen (z. B. Verpächter:innen), haben die Möglichkeit, einige der bodenbiodiversitätsfreundlichen Maßnahmen vorzuschreiben bzw. besonders problematische Praktiken auf den ihnen gehörenden Böden auszuschließen." (p. 919-920)

Again, no command, just an option: *haben die Möglichkeit*—"have the possibility." This keeps decision-making power with the reader, preventing science from appearing authoritarian.

Modern science communication strives to avoid alarmism—to inform without fear-mongering. This chapter follows that principle:

"Nevertheless, there can also be synergies. For example, humus-increasing measures in agriculture have positive effects on soil fertility, soil biodiversity, and at the same time on climate protection."^a

^aOriginal German: "Gleichwohl kann es auch Synergien geben. Beispielsweise haben humusmehrnde Maßnahmen in der Landwirtschaft positive Auswirkungen auf Bodenfruchtbarkeit, Bodenbiodiversität sowie gleichzeitig auf den Klimaschutz." (p. 920)

But is this always the best approach? The phrase *"there can also be synergies"* suggests that soil-friendly farming *might* help with climate change but avoids emphasizing its significance. At times, this neutrality becomes so vague that it stops being useful.

It Depends. But...

It depends. Every environmental science student says this constantly. Because nature isn't simple.

But is *"it depends"* the best way to communicate science? Scientists are **comfortable with uncertainty**. Policymakers? Farmers? The public? They need something **to hold onto**.

Perhaps we should reframe:

"It depends, but here's what we do know." "It depends, but this is what works best right now." "It depends, but we don't have to wait to act."

Sipping my coffee, I consider the work in front of me. So far everything is sound. And yet, as I turn the pages of this chapter, I find it all a bit... harmonious. Too smooth. Too neutral. Where is the messy, real-world friction that defines scientific progress?

Where's the Drama?

Science is not a peaceful agreement—it's a battlefield. Yet, this chapter reads as if everyone is in quiet consensus.

Where are the competing voices? The messy debates?

Competing interests shape soil biodiversity policy: farmers balance yield with conservation, policymakers juggle economic and ecological priorities, and agribusiness markets 'green solutions' that may serve profits more than the planet. Yet these tensions barely surface in the text.

The chapter acknowledges uncertainty, but what about controversy? Debate? Conflicting narratives? It lists opposing studies, but these counterpoints feel more like footnotes than real scientific contention. If the chapter stresses data gaps, why doesn't it reflect ongoing scientific disagreement?

"There is increasing evidence for synergies between climate mitigation, adaptation, and soil (biodiversity) conservation."^a

^aOriginal German: "Es gibt zunehmende Evidenz für Synergien zwischen Klimaschutz und -anpassung und Boden(biodiversitäts)schutz." (p. 920)

"Increasing evidence"—but from where? Large-scale field studies? Meta-analyses? Small-scale trials? Is dissenting research being acknowledged? The phrase is vague, leaving no way to assess confidence in the claim. Repeating *"it is generally recognized that..."* multiple times does not replace citations.

"Soil biodiversity is rarely considered in biodiversity assessments in Germany or Europe and is thus given little attention in current conservation policy."^a

^aOriginal German: *"Die Bodenbiodiversität wird sowohl in Deutschland als auch in Europa bisher kaum in Biodiversitätsgutachten berücksichtigt und findet daher nur wenig Beachtung in der aktuellen Naturschutzpolitik."* (p. 921)

But why is soil biodiversity overlooked? What competing narratives exist? Are sectors actively resisting its inclusion? Real conflicts shape soil biodiversity research and management:

- **Agriculture vs. Conservation**—Can large-scale farming and biodiversity truly coexist?^{2 3}
- **Regenerative Agriculture**—Is no-till farming always beneficial, or are trade-offs ignored?⁴
- **Industry Influence**—Are major corporations adopting genuine sustainability or just greenwashing?^{5 6 7}

These tensions are hinted at, but never fully engaged.

I glance up from my screen. Around me, my fellow students are lost in research papers, cross-referencing sources, ensuring every claim is airtight—because Reviewer #2 might find a gap.

We pride ourselves on accuracy, on credibility. And yet, in casual conversation, we often enough find ourselves repeating unverified claims.

We challenge misinformation, but we are not immune to cognitive shortcuts. Without realizing it, we build our own mythologies around science.

This chapter is careful—so careful—not to overstate. But in its fear of being wrong, does it miss the chance to counter what's already out there? It informs, but does it correct?

Where are the myths? Fighting Fiction

It's not enough to present facts, we need to fight the fiction. Thus, correcting misconceptions like these⁸:

- **"All earthworms are beneficial in agriculture"**—Some invasive earthworm species (e.g., *Amyntas* spp.) can disrupt soil ecosystems.
- **"Soil fungi are always beneficial for crops."** Some pathogenic fungi (e.g., *Fusarium*, *Phytophthora*) damage crops by infecting roots
- **"Soil biodiversity naturally recovers after farming stops."**—Soil biodiversity can take decades to recover.

So why not *call these misconceptions out directly*?

Instead, the chapter *plays it safe*, avoiding explicit confrontations with misinformation. And that's a huge missed opportunity.

The Missing Spectrum: More Than Just Data

Toward the end of the chapter, buried in technical discussion, is a brief nod to spirituality, symbolism, and aesthetics in soil science. An attempt—half-hearted at best—to acknowledge marginalized perspectives. It feels more like an afterthought than a meaningful engagement with alternative ways of understanding soil biodiversity.

Soil is not just a scientific issue; it is cultural, political, and deeply personal. Farmers, Indigenous communities, environmentalists, and industry leaders all relate to soil differently, shaping how they interpret scientific findings.

- For Indigenous communities, soil is a sacred, living entity. How does Western science engage with this worldview?
- For farmers, soil health is economic survival. Are their perspectives reflected in policy discussions?
- For urban environmentalists, soil is often invisible. How do we reconnect people with the ground beneath their feet?

²The Guardian, *Farm subsidies wrecked Europe's environment*.

³ResearchGate, *Conflicts between agriculture and biodiversity conservation in Europe*.

⁴MDPI Sustainability, *The regenerative agriculture debate*.

⁵Reuters, *Germany blocks oil firms from using past emissions credits to support biofuel industry*.

⁶Financial Times, *Corporate greenwashing in soil biodiversity claims*.

⁷ScienceDirect, *The role of industry in soil biodiversity narratives*.

⁸Wiley Online Library, *Misconceptions about biodiversity and soil health*.

These perspectives are mentioned but never fully explored. If science communication is meant to be inclusive and engaging, it cannot focus only on data and uncertainty—it must connect to real-world experiences.

I close the book and exhale. Around me, the café hums with quiet urgency—students scrambling to finish assignments, copy-pasting fluffy ChatGPT-generated sentences, fact-checking references to sound “scientific.” The irony isn’t lost on me. We chase precision, yet science is more than just facts.

I believe in the responsibility of science communication: to be transparent about uncertainties, honest about limitations, and rigorous in claims. This chapter does that. My reviewer’s eye catches minor flaws, places where clarity could improve. But on the whole, it follows the principles of good science communication.

And yet, something is missing.

Fascination. Lost in Translation.

The chapter explains *what we know and what we don’t*, but what about *why it matters*? Science isn’t just data and uncertainty—it’s a process of discovery, a way of understanding the world. And yet, this chapter remains clinical, missing the essential human element: **why we care**.

Facts alone won’t inspire action. People don’t just absorb knowledge; they engage with stories, with meaning. Science communication shouldn’t just provide stakeholders with options—it should offer them *new ways to think*. A shift in perspective, a spark of curiosity, is what drives change.

Where Are the Stories?

What if, instead of another section titled “*Soil Biodiversity: An Overview of Current Knowledge and Data Gaps*,” we had one called “*The Underground Architects: How Tiny Creatures Shape the World Beneath Our Feet*”?

What if, instead of dry explanations on “*Soil Microbial Interactions and Climate Resilience*,” we read about “*The Secret Lives of Microbes: The Billion-Year-Old Network Beneath Our Toes*”?

Scientific facts alone won’t change minds. **We act because something shifts in how we see the world.** If science communication invites curiosity, if it tells the stories that make knowledge come alive, then it has the power not just to inform, but to inspire.

Disclaimer: This is Bad Science Communication

I sit back in the café, skimming through this blog post, dissatisfied. The irony isn’t lost on me again — this piece, with all its catchy phrases and storytelling, isn’t just informing — it’s persuading. Am I convincing you, rather than letting you decide? I pretend certainty, but I question science communication every time I step in and out of academia. Where is the debate? The drama? The myths? What did I leave out?

There’s a rabbit hole to explore, but I won’t take you down it—not because I can’t, but because I, like many, am caught in the productivity trap. Or maybe I’m just lazy. And yes, parts of this post were written by ChatGPT.

So instead, here’s a biased, unbalanced view of what science communication can do, alongside other voices asking for better stories (go on, science communication police, call me out):

- *Étienne Fortier-Dubois*: Scientific Writing is a Miserable Grind & Crazy Science Idea: Tell People Something Is Possible When You Actually Have No Clue
- *Adam Mastroianni*: Is Psychology Going to Cincinnati? & There Are No Statistics in the Kingdom of God

And actually on the topic:

- *frontiers*: Protecting Soil Biodiversity: A Dirty Job, but Somebody’s Gotta Do It! from Soil Biodiversity Collection (yes this is for kids!)
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Back to the Soil, Back to the Message

I step outside, past the campus garden, where my classmates debate crop rotation, mud clinging to their boots. Tired, frustrated—yet still obsessed. Because they *feel* what they study matters.

That’s what science communication should do. Not just present facts, but reshape the way people see them.

We don’t just need to give people knowledge.

We need to give them a reason to want to know more.

References

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